



Lattice Boltzmann simulation of ion and electron transport during the discharge process in a randomly reconstructed porous electrode of a lithium-ion battery

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ABSTRACT

Rechargeable lithium-ion batteries (LIBs) have improved significantly over the past 20 years; however, the micro-scale working mechanism of these batteries, especially the influence of electrode microstructures on battery performance, still requires further study. The relationship between the electrode microstructure and battery performance is studied in detail by developing a two-dimensional (2D) lattice Boltzmann model to reveal the ion and electron transport within an LIB porous electrode. The present model is established by using randomly reconstructed electrode geometry and the coupling of the electrochemical reaction and ion and electron transport under complex boundary conditions. The influences of the micro-scale morphological features of the battery electrode on the local lithium concentration distribution, electric potential and macroscopic discharge performance are revealed. The randomness of the particle distribution and particle shape result in a heterogeneous distribution of the lithium concentration and electrode potential. In electrode particles, lithium exchanges are enhanced for small particles and improved at the edges and corners for large irregular particles. Regions with low lithium concentration produce a high electric potential. A large particle size and low porosity in an anode improve battery performance. Likewise, a small particle size and high porosity in a cathode also improve battery performance.

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1. Introduction

For the last two decades, lithium-ion batteries (LIB) have become an important power source for portable devices, such as cellular phones, laptops, and cordless power tools. Next-generation LIBs are utilized in large-scale applications, such as electric vehicles, power grids, and solar energy storage. However, capacity fading, insufficient power, and low energy density limit the performance of current LIB technology. Aside from these reliability issues, the electrode morphological features, which influence ion and electron transport, have also restricted the application of LIBs. A firm understanding of the electrode morphological properties and the relationship between microscopic transport mechanisms and macroscopic battery performance is essential to overcoming the aforementioned limitations of LIBs.

An LIB generally comprises a cathode material, an anode material, and a separator. Given the porous architecture of an electrode,

its electrochemical potential induces the lithium intercalation and deintercalation in the electrode material and drives the ion diffusion and migration within the electrolyte. The macroscopic working performance parameters of the battery, such as output power and cell voltage, influence the microscopic transport of the lithium ions and electrons. Two main factors, namely, material properties and electrode morphology, strongly influence the mechanism of transport of the lithium ion and electron in a porous electrode. Experimental investigations and numerical simulations have been conducted to evaluate the influence of this transport mechanism on cell performance. Experimental studies have shown that the morphological features, such as particle shape, particle size, and particle arrays, of the electrode material affect the cycling performance and battery capacity. As demonstrated in the experiment of Lu [1], smaller LiMn_2O_4 cathode particles with large interface area provide more lithium ions for diffusion and higher specific capacity. For MCMB anode particles, Zhang et al. [2] found that the first charge discharge efficiency increased from 86.2% to 90.5% when the particle size increased from 11.2 μm to 24.8 μm . Electrode microstructure directly affects the energy, power, lifetime, and reliability of LIBs. However, experimental investigations

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Nomenclature

c	discretized velocity direction	κ	ion conductivity (S cm^{-1})
c_s	lattice sound speed	σ	electron conductivity (S cm^{-1})
d	particle diameter (μm)	ϕ	electrical potential (V)
C	concentration (mol m^{-3})	Φ	interfacial electrostatic potential (V)
D	diffusivity ($\text{m}^2 \text{s}^{-1}$)	τ	relaxation time
F	Farady constant ($96,487 \text{ C mol}^{-1}$)	ω	weight factor
g	concentration distribution function	y	surface to maximum concentration ratio
h	potential distribution function		
H	height (μm)		
I	current density (A m^{-2})	Superscript	
l_0	length scale	<i>eq</i>	equilibrium
L	length (μm)	<i>max</i>	maximum
J	specially chosen constant in Eq. (14)	Subscript	
k	reaction rate in Eq. (6) ($\text{m}^{5/2} \text{s}^{-1} \text{mol}^{-1/2}$)	<i>a</i>	anode
K	specially chosen constant in Eq. (14)	<i>c</i>	cathode
n	normal vector	<i>e</i>	electrolyte
R	gas constant ($8.314 \text{ J mol}^{-1} \text{K}^{-1}$)	<i>g</i>	gas
s	surface area	<i>i</i>	direction of the lattice velocity
t	time	<i>k</i>	kth component
t_0	time scale	<i>l</i>	lattice unit
T	temperature (K)	<i>Li</i>	lithium ion
u	velocity vector	<i>p</i>	particle
Δx	space step	<i>s</i>	solid
Δt	time step	<i>Sep</i>	separator
α	reaction rate coefficient	<i>ohm</i>	ohmic
η	over-potential (V)		

have limitations because the morphological features of particles produced by chemical and physical preparations are stochastic and difficult to regulate. In addition, existing technology cannot reveal the uneven local lithium concentration and potential distribution induced by the morphology of the electrode. Therefore, the microscopic transport mechanism is difficult to study with experimental approaches; thus, the numerical method is preferred in studying the influence of the microscopic electrode morphology on the material transport and battery performance.

Numerous numerical methods have been proposed to improve the understanding of the transport phenomenon in porous electrodes as well as the relationship between microscopic transport and macroscopic cell output. Previous studies [3–5] initially proposed a mean-field pseudo two-dimensional (P2D) model that incorporates battery microstructure features into battery models. Microscopic effects were introduced into the battery model by homogenizing the spatial distribution of the electrode material and electrode porosity. Succeeding porous electrode models [6,7] utilized homogenized representations of the geometry and properties of the cathode and anode. These mean-field continuum models consider the simplified electrode geometry and the randomness of electrode particle size; however, particle shape and particle distribution are neglected. Consequently, the influence of electrode microstructure on the performance of LIBs has been ignored. The influence of the heterogeneity of porous electrode particles on battery performance can be described in numerical simulations by reconstructing an electrode with a geometry that is anisotropic in terms of physical properties as well as in electrode architecture.

To generate electrode geometry with different particles shapes, alternative approximations of electrode particle have been utilized. Most studies regarding electrode microstructure have utilized regularly shaped particles to represent the heterogeneity of electrode particles. In the previous work of present authors, the circular shape particles are utilized to reconstruct the electrode geometry and the effect of the particle morphological effect on the battery performance is investigated [8]. Moreover, García et al. [9] used

randomly distributed circular particles to describe the spatial variations in the electrochemical fields of porous electrode microstructures. The utilization of randomly distributed particles is preferred to the mean-field method because the former can reveal microscopic physical processes that cannot be observed in mean-field porous electrode models. Less et al. [10] used three different generic electrode geometries with cubic particles in their simulation and proposed the correlation of nonhomogeneous material properties with macroscopic battery performance. Wang and Sastry [11] established a three dimensional (3D) model of a porous cathode in a lithium-ion polymer cell. These numerical studies assumed that the shape of particles in an active electrode material is either spherical or cubic; however, recent investigations on cell characterization have demonstrated that the morphology of the particles differs from the assumed spherical or cubic shape [12,13].

Irregular electrode architecture should be constructed, and the effects of the randomness of shape, morphology, and packaging of the electrode particles can also be simulated to reflect the realistic morphology of electrode microstructure. Several attempts have been made to utilize realistic electrode architecture. Smith et al. [14] utilized the real LiCoO_2 battery cross-section of an LIB as the modelling geometry in their investigation. Chung et al. [15] conducted a 3D multiphysics simulation using a 3D particle model reconstructed from Atomic Force Microscope scan data. Wiedemann et al. [16] developed the microstructural geometry of an LIB cathode that was reconstructed based on focused-ion-beam experiments. Investigations utilizing the electrode geometry with random particle shapes and distribution have predicted battery performance more precisely than those using the regular spherical or cubic particle geometry. In numerical investigations, the finite element method (FEM) [8–10] and finite volume method (FVM) [15,16] have often been utilized to solve the charge and species transport in battery electrochemical systems. However, building a mesh for irregular electrode particles in the FEM or FVM is challenging, and considerable computational resources are consumed in the process.

The lattice Boltzmann method (LBM) is a promising approach to investigate the complex transport processes with stochastic electrode geometries because of its excellent numerical stability and constitutive versatility [17–19]. Recently, the LBM has been widely utilized to study batteries, such as fuel cells, polymer electrolyte membranes (PEMs), and flow batteries while considering the reactant transport and electrochemical reactions at the interface. Akinaga et al. [20] conducted a lattice Boltzmann simulation of the proton transport within a Nafion channel and explained the proton diffusion within a highly disordered porous structure. Qiu et al. [21] presented a 3D model of the electrochemical processes in a vanadium redox flow battery. Tabe et al. [22] conducted numerical simulations by using the LBM to elucidate the gas flow in the gas diffusion layer of a PEM fuel cell. Given the successful utilization of the LBM in studying different batteries, the LBM has also been used to investigate LIBs. Several studies on LIBs have been conducted to investigate the electrolyte transport and determine the effective transport coefficient. Lee et al. [23] simulated the electrolyte transport in the porous electrode structure of an LIB. The complex microscopic transport mechanisms of the electrolyte in the porous electrode were investigated, but the electrochemical reaction on the electrode was ignored. To understand the effect of the electrode microstructure on the performance of an LIB, Wu et al. [24] reconstructed a LiCoO_2 cathode and calculated the effective transport coefficient of the electrolyte (or solid) phase. The geometric connectivity and tortuosity of specific phases were obtained; however, the influences of these morphological features on battery performance were not discussed. In general, LBM has been utilized in the research on LIBs, but only the evaluation of the porous structure and transport properties of these batteries have been emphasized. Investigations that incorporate ion and electron transport and electrochemical reaction are few. As stated in previous investigations on batteries [20,22,23], 2D modelling is commonly utilized due to its high efficiency and reliability. Moreover, considering the tremendous computing workload and poor convergence in 3D modelling, it is rational to build 2D LBM model to solve the coupling mass transport in porous electrode.

A 2D framework for the LBM simulation is proposed for the species and charge transport in LIB porous electrodes. To evaluate the influences of the heterogeneous distribution and random size and shape of particles on battery performance, a stochastic method, called the quartet structure generation set (QSGS), is utilized to generate the electrode geometry. The reconstructed battery electrode comprises active cathode and anode particles. Electrodes with different particle sizes, particle shapes, and porosity values are considered. The effects of the electrode microstructure on the species and charge spatial variation and battery performance are investigated.

2. Physical problem description and mathematical model

2.1. Schematic of the battery working process

A schematic of a typical LIB is shown in Fig. 1. The LIB used in this study comprises cathode particles, anode particles, a separator, a negative current collector, and a positive current collector. The electrode particles are aggregated by a binder, and they form the porous structure of the electrode. The void spaces in the porous structure are filled with electrolytes. The active materials of the cathode particles and anode particles are lithium manganese oxide ($\text{Li}_y\text{Mn}_2\text{O}_4$) and graphite, respectively. The height of the battery is 126 μm . The thicknesses of the cathode, separator, and anode are 120, 30, and 75 μm , respectively. Five typical regions are selected to show clearly the boundary conditions and equations for the battery. These regions are the electrolyte phase (I), electrode particles

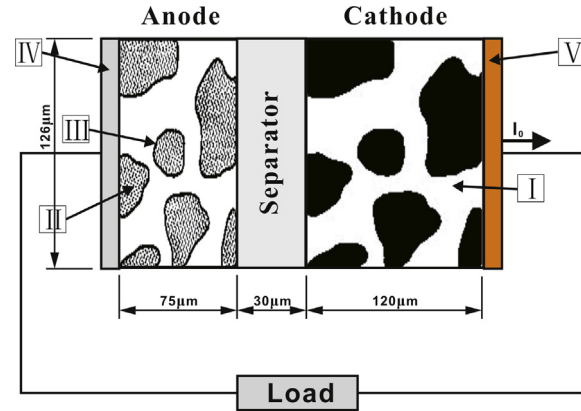
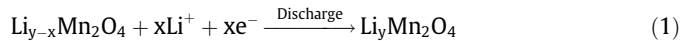


Fig. 1. Schematic of a typical rechargeable lithium-ion battery. (I) Electrolyte phase; (II) active material particles; (III) reactive interface; (IV) anode current collector; (V) cathode current collector.

(II), solid electrolyte interface (III), anode current collector (IV), and cathode current collector (V). The discharge process is selected for the evaluation of the lithium transport and electrochemical reaction. Lithium is driven by the electrochemical potential via diffusion and migration and is transferred from the anode to the cathode in the electrolyte phase (I). A charge reconstruction process occurs at the solid electrolyte interface (III), where lithium depletion and lithium accumulation occur. Pure ion diffusion is assumed within the particles, as shown in (II) and (III). The current densities I_0 and $-I_0$ are imposed on interfaces (IV) and (V), respectively. The battery is discharged at 1C-rate and the applied current density is calculated with respect to the theoretical capacity of the battery. The electrochemical reactions in the two electrodes are written as follows:



2.2. Mathematical formulation for the battery domain

In the battery, the ion transport, electron transport, and electrochemical reaction are the basic physical processes considered in the modelling. In previous mean-field P2D models, the transport equations for the ions and electrons in the solid phase and electrolyte phase are solved in each node within an evenly distributed calculation domain. The influence of the electrochemical reaction is introduced into the governing equations by adding a source term related to the charge transfer rate. In the present model, the transport equations for the ions and electrons in both phases are derived separately. Moreover, the electrochemical charge transfer rate and the interfacial electrostatic potential are set at the solid electrolyte interface and utilized as the boundary conditions of the equations for the ion and electron transport, respectively. Therefore, the morphological properties of the electrode particles can be retained.

The lithium-ion transport in the electrode particles can be defined as follows:

$$\frac{\partial C_s}{\partial t} - \nabla \cdot (D_s \nabla C_s) = 0 \quad (3)$$

Whereas the transport of the lithium ion in the electrolyte can be expressed as follows:

$$\frac{\partial C_e}{\partial t} - \nabla \cdot (D_e \nabla C_e) - \nabla \cdot \left(\frac{D_e z F C_e}{RT} \nabla \phi_e \right) = 0 \quad (4)$$

where C_s is the lithium-ion concentration in the solid particles, and D_s is the diffusivity of lithium in the solid particles, C_e is the lithium-ion concentration in the electrolyte, and D_e is the diffusivity of lithium in the electrolyte, ϕ_e is the electrolyte voltage, F is the Faraday constant, z is the charge number of the ions, R is the gas constant, and T is the temperature.

The electron transport in the electrode particles is as follows:

$$\nabla \cdot (\sigma \nabla \phi_s) = 0 \quad (5)$$

Whereas the electron transport in the electrolyte can be expressed as follows:

$$\nabla \cdot \kappa \nabla \phi_e + \nabla \cdot \left(\frac{D_e z F C_e}{RT} \nabla C_e \right) = 0 \quad (6)$$

where ϕ_s is the solid-phase voltage, σ is the solid-phase electrical conductivity, κ is the electrolyte ion conductivity. The physical properties referred to in the control equations for the battery are summarized in Table 1.

2.3. Electrochemical kinetics

The electrochemical charge transfer rate is controlled by the Butler–Volmer relation [3,4]:

$$I = i_0 [e^{(\alpha_a F/RT)\eta} - e^{-(\alpha_c F/RT)\eta}] \quad (7)$$

The exchange current density is calculated by the following equation:

$$i_0 = F k_r C_e^{\alpha_a} (C_{\max} - C_{\text{surf}})^{\alpha_a} C_{\text{surf}}^{\alpha_c} \quad (8)$$

where k_r is the reaction rate constant; $C_{\max}$ and C_{surf} are the saturated lithium concentration and surface concentration in the solid particle; α_a is the anodic reaction rate coefficient; and α_c is cathodic reaction rate coefficient. The lithium exchange at the solid electrolyte interface is governed by Eqs. (7) and (8). The interfacial overpotential η indicates the deviation from the equilibrium potential.

In the two dimensional geometry, the anode current collector and the cathode current collector are not perfectly connected by the particle, since many isolated particles occur. Therefore, the potential distribution in the solid phase induced by the applied current is hard to be calculated. In order to obtain the electric potential evolution in the isolated particles, a simplified approach proposed by García [9,14] is employed in this study. By utilizing

this approach, the accurate local voltage distributions can be reserved even through some in-plane connection details of electrode geometry are missing. An interfacial electrostatic potential Φ is asserted on the particle surface and is utilized as the boundary condition for the electric potential distribution in particles. Φ is calculated by the following equation:

$$\Phi = U(y) \pm \eta \quad (9)$$

where $U(y)$ is the lithium-dependent equilibrium potential and is introduced to couple the ion transport and electron transport. The potential data in this study are obtained from previous study [4]. At the reactive interface, $U(y)$ is denoted as the electric potential boundary for the electrolyte. The calculation of $U(y)$ is as follows:

The following equation applies to graphite particles:

$$U(y) = 0.7222 + 0.1387 \times y + 0.029 \times y^{0.5} - 0.0172 \times y^{-1} \\ + 0.0019 \times y^{-1.5} + 0.2808 \times e^{(0.9-15 \times y)} \\ - 0.7984 \times e^{(0.4465 \times y - 0.4108)} \quad (10)$$

The following equation applies to $\text{Li}_x\text{Mn}_2\text{O}_4$ particles:

$$U(y) = 4.1983 + 0.0566 \times \tan(-14.5546 \times y + 8.6094) - 0.02754 \\ \times ((0.9984 - y)^{-0.4925} - 1.9011) - 0.1571 \times e^{(-0.04738 \times y^8)} \\ + 0.8102 \times e^{(-40 \times (y - 0.1339))} \quad (11)$$

where $y = C_{\text{Li}}/C_{\text{Li}}^{\max}$ is the ratio of the particle surface concentration to the total concentration.

The interfacial overpotential is denoted by $\eta \approx R_{\text{ohm}} I$. The current density at the electrode–electrolyte interface is denoted by I , and R_{ohm} is the contact interfacial ohmic resistance between particles and electrolyte. As indicated in Ref. [9], the value of the interfacial resistance is fitted from experimental discharge curves. The same values of R_{ohm} for anode and cathode with Ref. [9] are adopted in this study, since the cathode material and electrolyte solute utilized in this study are the same with that used in the reference. The cathode electrode contact resistance is set to $1.2 \times 10^{-2} \Omega$, and the anode electrode contact resistance is set to $2 \times 10^{-3} \Omega$.

2.4. Initial and boundary conditions

The initial and boundary conditions for the electrochemical model of the LIB are summarized in Table 2. The galvanostatic dis-

Table 1
Material properties and initial conditions utilized in the simulation [8,9].

Parameter	Name	Value	Unit
D_a	Lithium diffusivity in the anode Li_yC_6 particles	3.9×10^{-14}	$\text{m}^2 \text{s}^{-1}$
D_c	Lithium diffusivity in the cathode $\text{Li}_x\text{Mn}_2\text{O}_4$ particles	1×10^{-13}	$\text{m}^2 \text{s}^{-1}$
D_e	Salt diffusivity in the electrolyte	7.5×10^{-11}	$\text{m}^2 \text{s}^{-1}$
$C_c^{\max}$	Maximum solid-phase concentration in the cathode	22,860	mol m^{-3}
$C_a^{\max}$	Maximum solid-phase concentration in the anode	26,390	mol m^{-3}
C_c^0	Initial solid-phase concentration in the cathode	3900	mol m^{-3}
C_a^0	Initial solid-phase concentration in the anode	14,870	mol m^{-3}
C_e^0	Initial electrolyte salt concentration	2000	mol m^{-3}
σ_a	Electric conductivity of Li_yC_6 particles	1×10^{-2}	S cm^{-1}
κ	Ion conductivity of the electrolyte	2.53×10^{-2}	S cm^{-1}
σ_c	Electric conductivity of the $\text{Li}_x\text{Mn}_2\text{O}_4$ particles	3.8×10^{-2}	S cm^{-1}
$k_{r,\text{neg}}$	Negative reaction rate constant	1.9×10^{-9}	$\text{m}^{5/2} \text{s}^{-1} \text{mol}^{-1/2}$
$k_{r,\text{pos}}$	Positive reaction rate constant	1.9×10^{-9}	$\text{m}^{5/2} \text{s}^{-1} \text{mol}^{-1/2}$
α_a	Anodic reaction rate coefficient	0.5	
α_c	Cathodic reaction rate coefficient	0.5	
L_a	Length of the anode	75	μm
L_{sep}	Length of the separator	30	μm
L_c	Length of the cathode	120	μm
L	Length of the calculation domain	225	μm
H	Height of the calculation domain	126	μm
I_0	Applied current density	12.5	A m^{-2}

Table 2

Initial and boundary conditions for the electrochemical model of the lithium-ion battery.

Governing equations	Initial conditions	Boundary conditions
Electron transport in the electrolyte		$-\sigma \frac{\partial \Phi_i}{\partial x} \Big _{x=0} = \sigma \frac{\partial \Phi_i}{\partial x} \Big _{x=L} = I_0$, $\frac{\partial \Phi_i}{\partial y} \Big _{y=0} = \frac{\partial \Phi_i}{\partial y} \Big _{y=H} = 0$
Lithium-ion transport in the electrolyte	$C_e = C_e^0$	$\frac{\partial C_e}{\partial x} \Big _{x=0} = \frac{\partial C_e}{\partial x} \Big _{x=L} = 0$, $\frac{\partial C_e}{\partial y} \Big _{y=0} = \frac{\partial C_e}{\partial y} \Big _{y=H} = 0$
Species at the solid–electrolyte interface		$D \frac{\partial C}{\partial n} = \frac{J}{2F}$
Electron at the solid–electrolyte interface		$\Phi = U(y) \pm R_{ohm} I$

charge mode is applied to the solid charge conservation equation at the left and right boundaries, and the no charge flux is applied to the liquid charge conservation equation at the top and bottom boundaries of the boundary condition. For the species conservation equations, the no species flux condition is applied to the electrolyte. In addition, the cutoff voltage for the battery discharge is set to 3 V.

3. Lattice Boltzmann model for the electrochemistry and geometry reconstruction

3.1. Lattice Boltzmann model for the battery

The LBM is utilized to solve the governing equations of the transport of ions and electrons. In an LIB, however, the flow of the electrolyte is ignored, and the battery can be viewed as a diffusion–reaction system. The D2Q5 model is utilized for the discrete velocity because this model considerably reduces computational expense and is sufficiently accurate in a 2D diffusion case [25,26].

The following evolution equation is used to describe the ion transport in the solid phase [19]:

$$g_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) - g_i(\mathbf{x}, t) = -\frac{1}{\tau_D} [g_i(\mathbf{x}, t) - g_i^{eq}(\mathbf{x}, t)] \quad (12)$$

where $g_i(\mathbf{x}, t)$ is the distribution function of specie concentration in the direction i at lattice site $\mathbf{x}$ and time t ; $\mathbf{x}$ is the location vector; g_i^{eq} is the corresponding equilibrium function; τ_D is a relaxation time; and $\mathbf{c}_i$ is the discretized velocity direction and is calculated as:

$$\mathbf{c}_i = \begin{cases} (0, 0) & i = 0 \\ (\cos \theta_i, \sin \theta_i) c_s, \theta_i = (i-1)\pi/2, & i = 1-4 \end{cases} \quad (13)$$

where c_s is the lattice sound speed.

According to Eq. (4), an ion migration is considered in the electrolyte; thus, an electric force term is added to the evolution equation for the ion transport within the electrolyte [19], the evolution equation is written as follows:

$$g_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) - g_i(\mathbf{x}, t) = -\frac{1}{\tau_D} [g_i(\mathbf{x}, t) - g_i^{eq}(\mathbf{x}, t)] + \frac{zF(\mathbf{c}_i - \mathbf{u}) \cdot \nabla \phi}{RT} g_i^{eq}(\mathbf{x}, t) \quad (14)$$

where $\mathbf{u}$ is the velocity vector; g_i^{eq} is usually calculated as follows:

$$g_i^{eq} = C(J_i + K_i \mathbf{c}_i \cdot \mathbf{u}) \quad (15)$$

where C is the concentration of the specie, and K_i is a constant equal to 1/2 in a 2D simulation. By the Chapman–Enskog expansion, the lattice Boltzmann equations, Eqs. (12) and (14), for the ion transport can be reverted to macroscopic equations, Eqs. (3) and (4), respectively.

In Eq. (15), J_i is given by the following:

$$J_i = \begin{cases} J_0, & i = 0 \\ (1 - J_0)/4, & i = 1, 2, 3, 4 \end{cases} \quad (16)$$

where J_0 can be a value from 0 to 1 according to the diffusivity, D . The concentration of the specie C is calculated as follows:

$$C = \sum g_i \quad (17)$$

The diffusivity in the lattice unit is related to the collision time and is obtained by the following:

$$D = \frac{1}{2} (1 - J_0) (\tau_D - 0.5) \frac{\Delta x^2}{\Delta t} \quad (18)$$

With regard to the electron transport, the following evolution equation is used to describe the electron transport in the solid phase:

$$h_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) - h_i(\mathbf{x}, t) = -\frac{1}{\tau_h} [h_i(\mathbf{x}, t) - h_i^{eq}(\mathbf{x}, t)] \quad (19)$$

With the electron conduction in the electrolyte considered, the evolution equation is written as follows:

$$h_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) - h_i(\mathbf{x}, t) = -\frac{1}{\tau_h} [h_i(\mathbf{x}, t) - h_i^{eq}(\mathbf{x}, t)] + \omega_i \frac{F \nabla \cdot (z D \nabla C_{Li})}{RT} \quad (20)$$

where h_i is the function of the electric potential distribution in the direction i , ω_i is the weigh coefficient, for which $\omega_0 = 1/3$ and $\omega_{1-4} = 1/6$ for the D2Q5 model; and h_i^{eq} is the corresponding equilibrium function, which is often calculated as follows:

$$h_i^{eq} = \begin{cases} K_0 \phi & i = 0 \\ \frac{(1-K_0)\phi}{4} & i = 1, 2, 3, 4 \end{cases} \quad (21)$$

where K_0 has a value from 0 to 1 according to the electron conductivity; ϕ is the electric potential.

The electric conductivity in the lattice unit is related to the collision time and is obtained by the following:

$$\sigma = \frac{1}{2} (1 - K_0) (\tau_h - 0.5) \frac{\Delta x^2}{\Delta t} \quad (22)$$

By the Chapman–Enskog expansion, the lattice Boltzmann equations, Eqs. (19) and (20), for the electron transport can be reverted to macroscopic equations, Eqs. (5) and (6), respectively. The electric potential can be obtained as follows:

$$\phi = \sum h_i \quad (23)$$

3.2. Boundary condition treatment

Attention should be paid to the treatment of the interface where the electrochemical reactions occur. For lithium–ion transport, the ion flux across the reaction boundary can be derived from the current density. The value of g_4 , g_1 , and g_3 can be obtained from the neighbor nodes after each streaming process; the unknown distribution function, g_2 , is calculated as follows [25]:

$$g_2 - g_4 = -\frac{D}{c_s} \frac{\partial C}{\partial n} \quad (24)$$

where D is the diffusivity of the solute specie.

Given that the nonequilibrium part of the distribution function has opposite signs in opposite directions, the sum of the two distribution functions in the two opposite directions equals the sum of the corresponding equilibrium functions.

$$g_2 + g_4 = g_2^{eq} + g_2^{neq} + g_4^{eq} + g_4^{neq} = g_2^{eq} + g_4^{eq} \quad (25)$$

where g_2^{neq} and g_4^{neq} are the nonequilibrium parts of the corresponding distribution function.

The following equations can be obtained by combining the boundary conditions expressed in Eqs. (7), (15), (24), (25):

$$g_2 + g_4 = \frac{1 - J_0}{2} C \quad (26)$$

$$g_2 - g_4 = -\frac{I}{zF} \quad (27)$$

Subsequently, the unknown distribution function, g_2 , which enters the electrolyte domain, can be calculated and the lithium concentration at the reaction surface can be obtained.

The physical time step and physical diffusion coefficient are determined through the following dimensionless relation [27]:

$$\left(\frac{D\Delta t}{\Delta x^2}\right)_{LBM} = \left(\frac{D\Delta t}{\Delta x^2}\right)_{REAL} \quad (28)$$

Eq. (28) shows that the physical time step and physical diffusion coefficient are controlled by the grid resolution and relaxation parameter. Three diffusion processes are considered in this study, namely, the ion diffusion in the electrolyte, the lithium diffusion in the anode particles, and the lithium diffusion in the cathode particles. Table 1 shows that the diffusivity values in the three diffusion processes are different. Different lattice diffusivity values should be assigned to reflect the different diffusion speeds in the three diffusion processes. Therefore, the lattice diffusivity, D_{el} , of 0.7 is selected for the salt diffusion in the electrolyte, and the characteristic time step is $\Delta t = 3.3 \times 10^{-4}$ s. Given these values, the diffusivity values for the lithium diffusion in the anode particles and in the cathode particles can be calculated. The corresponding dimensionless lattice units for the physical units are summarized in Table 3.

3.3. Construction of the porous electrode geometry

The irregular porous structures are generated by computerized image reconstruction method QSGS [28], which is a convenient and efficient method to identify different porous characteristics. The particle size and porosity of electrode are most concerned factors affecting the battery performance. As the quantitative regulation of the two factors is very difficult for the experiment with existing technologies, QSGS method is therefore adopted due to its feasibility of generating porous structures with alterable particle size and porosity. Under this circumstance, the utilization of QSGS is necessary even though it sacrifice some accuracy compared with tomography data. In QSGS, four parameters, namely, core distribution probability, directional growth probability, phase interaction growth probability, and phase volume fraction, are identified to control the internal morphological features of the granular media. The porosity and particle size are controlled by the phase volume fraction and core distribution probability, respectively. The generation procedure of the porous media follows the steps below:

- Stochastically locate the particle cores based on the core distribution probability in a grid system. Then, each node in the grid will be assigned a random number by a uniform distribution function within (0, 1). A node will be determined as the core if its random number is no greater than the core distribution probability.
- Grow each element of particles to its neighboring nodes in all directions based on give directional growth probability. New random numbers will be assigned to neighboring nodes of new grown element. The neighboring node will become part of the growing phase if its random number is no greater than directional growth probability.
- Repeat growing process of (b) until the volume fraction reaches the expected value.

Fig. 2(a) shows a binary image of the porous electrode structure generated based on these conditions. As depicted in Fig. 2(b), the electrode particle phase and the void phase are characterized as “1” and “0,” respectively. Subsequently, the universal boundary recognition and classification program (BRCP) [29] for porous media is employed. As shown in Fig. 2(c), 14 types of boundary nodes are considered. Flat wall nodes (B1–B4) are within a straight-line boundary. Convex nodes (B5–B8) and concave nodes (B9–B12) are observed at the points of intersection between two straight-line boundaries. Convex-to-convex nodes (B13 and B14) represent the intersection points of four straight-line boundaries.

The porous electrode structures are generated by using the established porous electrode generation method and by inputting the porosity and core distribution probability. The calculation domains are 1196×670 lattice units in size. The structure of the irregular particles is shown in Fig. 2(d). The particles generated by this stochastic method are irregular in shape and have a random distribution. Compared with the spherical particle structure, the morphological structure of the irregular particles approximates the realistic electrode architecture obtained from the SEM images. During the structure construction, the separator part is ignored, and the porosity is controlled by the area ratio of the solid phase in the anode or cathode area. The porosity is considered, and a structure with porosity values, ε , 0.4, 0.5, and 0.6 are generated for both the cathode and anode. In order to validate the rationality of the generated irregular structure, the pore characteristic of generated structure is compared with that of the SEM data. The SEM figure of electrode cross-section is obtained from Ref. [23]. The pore distributions of cathode region in two structures, QSGS and SEM, are calculated. As shown in Fig. 3, the pore distributions have shown similar variation trend. Most of the pore size values fall into the range of 20–100 μm for two structures. The porous generation step is also utilized to generate electrode structures with different particle sizes. As in previous studies [17,29], the average particle size is utilized to represent the irregularity in the particle size in the porous electrode. The average particle size, d_p , is calculated as follows:

$$d_p = 6/s_v \quad (29)$$

where s_v is the specific surface area, that is, the ratio of the total length of all boundary lines to the total area of the domain. The

Table 3
Physical parameters and corresponding dimensionless lattice units for the simulation.

Parameter	Physical symbol	Physical value	Lattice symbol	Lattice value
Calculation domain length	L	2.25×10^{-4} m	L_l	1196
Calculation domain height	H	1.26×10^{-4} m	H_l	670
Grid step	$\Delta x(\Delta y)$	1.88×10^{-7} m	$\Delta x_l(\Delta y_l)$	1
Lithium diffusivity in the anode particles	D_a	3.9×10^{-14} m ² s ⁻¹	D_{al}	1/2700
Lithium diffusivity in the cathode particles	D_c	1×10^{-13} m ² s ⁻¹	D_{cl}	1/1070
Salt diffusivity in the electrolyte	D_e	7.5×10^{-11} m ² s ⁻¹	D_{el}	0.7

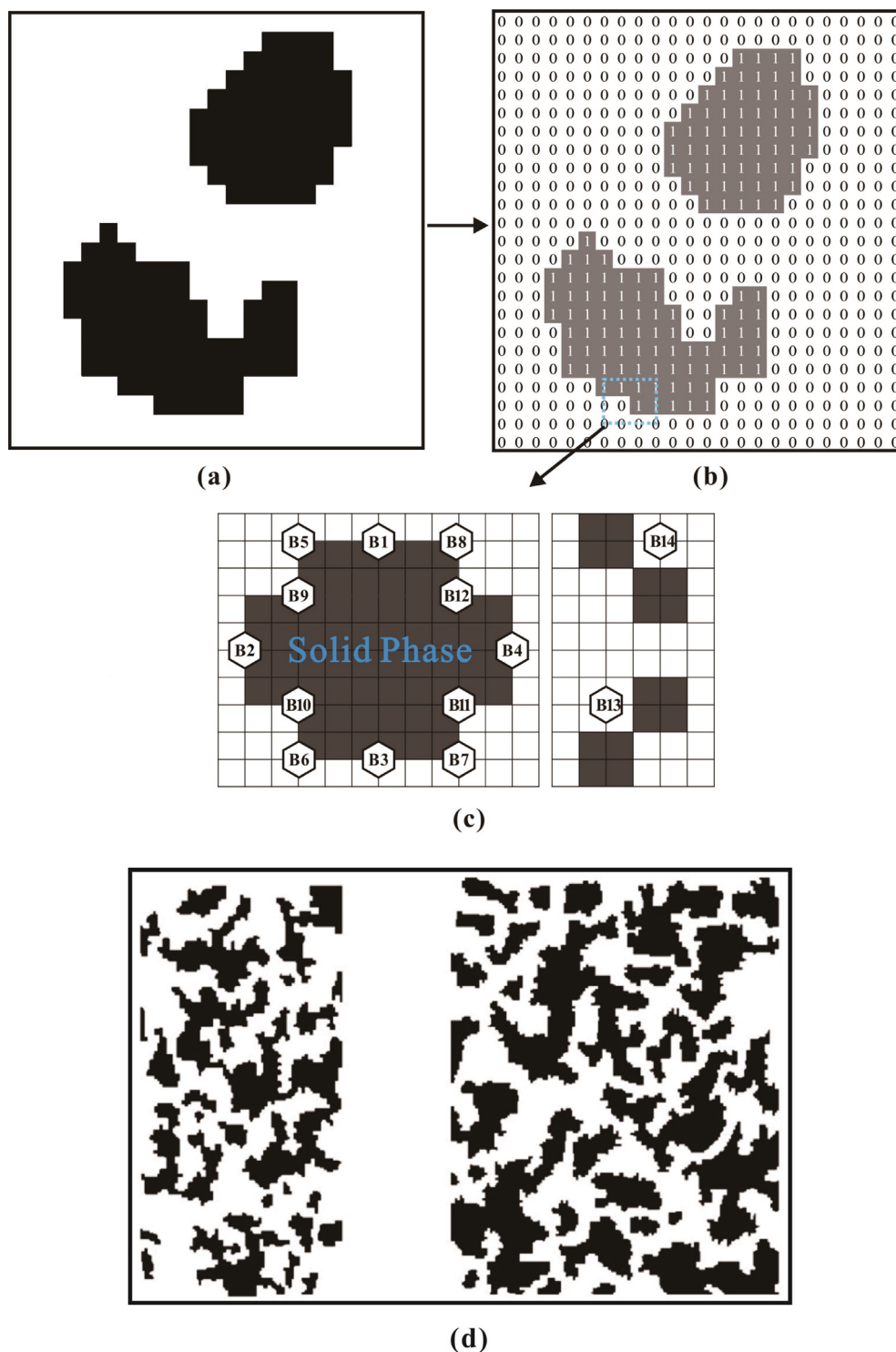


Fig. 2. Construction of the porous geometry. (a) Irregular geometry; (b) phase functions of the solid phase and void phase; (c) illustration of 14 types of boundary nodes; (d) electrode geometry with irregular particles.

morphological features of the reconstructed porous structures in the two groups are listed in Table 4.

4. Results and discussion

4.1. Model validation

The simulation results of the present model are compared with those of a P2D model. The voltage responses predicted by the pre-

sent LBM are compared with those obtained utilizing the commercial software COMSOL Multiphysics. In the COMSOL simulation, the FEM-based P2D model is constructed employing the method proposed by Cai and White [30]. Moreover, the electrode material, initial conditions, and boundary conditions have the same values between the LBM simulation and the COMSOL simulation. As shown in Fig. 4, the voltage responses variations with respect of depth of discharge (DOD) of LBM and the COMSOL simulation results are compared. The result of LBM model agree well with that

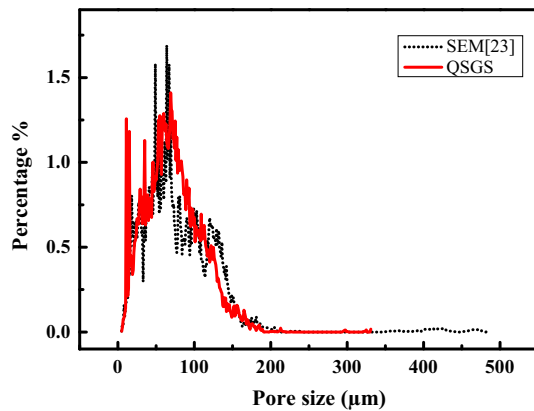


Fig. 3. Pore sizes distributions in QSGS and SEM.

Table 4

Structure parameters for reconstructed porous media with different average particle sizes and porosity values.

	Porosity	Average particle size (μm)	Porosity	Average particle size (μm)
Cathode	0.6	16.58	0.6	23.69
	0.6	28.97	0.5	23.65
	0.6	38.53	0.4	23.66
Anode	0.6	17.48	0.6	22.39
	0.6	27.07	0.5	22.28
	0.6	37.76	0.4	22.85

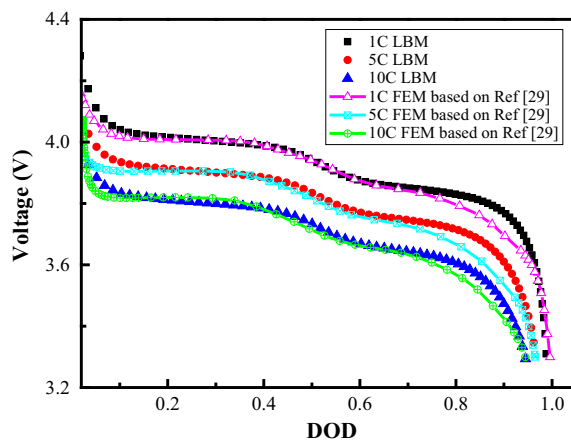


Fig. 4. Voltage responses at different discharge rates.

of FEM model when $0.1 < \text{DOD} < 0.8$. Little divergences occur at first and last stage of the discharge ($\text{DOD} < 0.1$ and $\text{DOD} > 0.8$). This is mainly induced by the difference of material properties used in the LBM model and FEM model. In the FEM model, the porosity-related physical properties, such as diffusion coefficient and electrical conductivity, are used. Comparatively, the real physical properties are used in the present pore scale model. The divergences in Fig. 4 imply that physical properties are more sensitive at the early stage ($\text{DOD} < 0.1$) and last stage ($\text{DOD} > 0.8$). The differences of transport properties result in the difference in lithium concentration change in electrode particles for two models and finally lead to divergences in voltage.

4.2. Lithium concentration distribution

The irregular randomness of the electrode particles induces an inhomogeneous distribution of lithium in the electrode particles

and in the electrolyte. The anisotropic features of the lithium concentration distribution can be analyzed by studying the time evolution of the lithium concentration in the battery electrode with irregular geometry (Fig. 5(a)). The lithium concentration distribution is normalized to the solubility limit of lithium in the anode particles, i.e., $26,390 \text{ mol m}^{-3}$. As the discharge process continues, the lithium concentration decreases in the anode particles but increases in the cathode particles. These contrasting variations are caused by the electrochemical reaction occurring at the reactive interface. As a result of the electrochemical reaction, lithium deintercalates from the anode particles but intercalates into the cathode particles. Meanwhile, the electrochemical potential drives the lithium in the electrolyte to move from the anode to the cathode. In addition, after 1320 s, several small graphite particles are completely depleted of lithium, as shown by the dark area. To some extent, the electrode microscopic properties account for the macroscopic feature of the battery. After depletion, the inactive particles increase the resistance of the cell and diminish the delivered instantaneous power density at the last stage of the discharge process.

The detailed lithium distribution properties and the pattern of the lithium concentration variation can be determined using the morphological features of the electrode particles by first obtaining the lithium concentration distribution of the cathode of the electrode at 1980s, as shown in Fig. 5(b). The cathode is divided into two regions according to the particle aggregation density and the distance to the separator. Region 1 is located near the separator and has a low particle density. By contrast, region 2 is located far from the separator and has a high particle density. Both particle aggregation and particle location affect the lithium concentration in the electrolyte. As depicted in Fig. 5(b), a remarkable lithium concentration gradient occurs in the electrolyte. The electrolyte lithium concentration in region 2 is significantly lower than that in region 1. This result implies that a lower electrolyte lithium concentration is produced for the cathode in the region located far from the separator and with intensive particle distribution. This outcome is attributed to three mechanisms. First, the diffusion of lithium from the separator to the cathode particles that are located far from the separator is difficult because the diffusion path is long. Second, the high particle density improves the lithium consumption in the electrolyte. Third, for the region with high particle density, the diffusion path of lithium is likely to be blocked. By contrast, for the anode, a lower electrolyte lithium concentration is achieved in the region located near the separator with a sparse particle distribution. The electrolyte lithium gradient in the anode is not as remarkable as that in the cathode. The morphology of the electrode particles should be monodispersed and uniformly distributed to reduce the lithium gradient in the electrolyte, as demonstrated in the work of García [9,14].

For the investigation of the detailed concentration change in the electrode particles, two regions, i.e., regions α and β , are selected in the anode and cathode, as illustrated in Fig. 6(a). The enlarged region α shows that the lithium concentration in small anode particles is lower than that in the large anode particles. In addition, the local lithium depletion is improved at the edges and corners of large anode particles. On the contrary, the enlarged part of region β shows that the lithium concentration in the small cathode particles is higher than that in large particles and that the lithium accumulation is enhanced at the edges and corners of large cathode particles. Two mechanisms can explain this outcome. First, the lithium concentration change is difficult in large particles with a strong lithium storage capacity. Second, the rough surface at the corner and edge sites of the large particles improves the effective reactive area; consequently, the lithium exchange rate is improved and faster depletion and accumulation rates are induced.

The lithium concentration changes at the edges and center of the anode and cathode particles are illustrated to quantify the

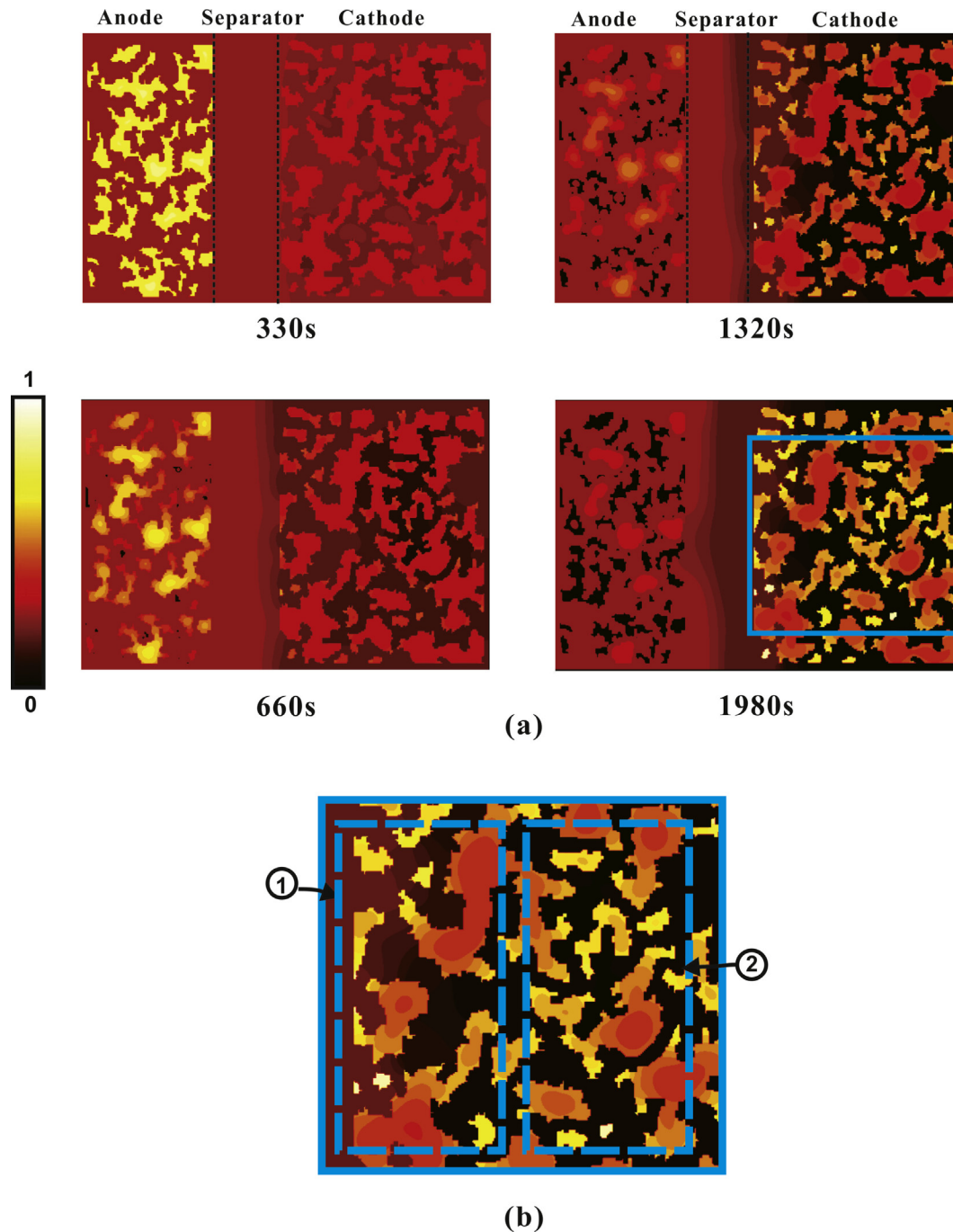


Fig. 5. Normalized lithium concentration distributions in the solid phase and electrolyte phase at different times for the electrode with irregular particle geometry. (a) Evolution of lithium concentration; (b) lithium concentration distribution of a segment in the cathode.

lithium concentration change in the different areas of a large electrode particle. As shown in Fig. 6(a), the lithium concentrations at edge point A and center point B are for the anode, and the lithium concentrations at edge point C and center point D are for the cathode. As shown in Fig. 6(b), the anode lithium concentration at edge point A continuously decreases during the discharge process, whereas a remarkable lag at center point B is identified before 297 s. After this “plateau” state, the concentration at center point B begins to drop at a speed similar to that at edge point A to form a constant concentration difference between these two points. This outcome occurs because of the diffusion of lithium from the edge point to the center point in the electrode particles. As shown in Fig. 6(c), a similar lag in lithium increases at center point D in

the cathode particles can be found before 198 s. The lithium concentration at points C and D both increase at a constant speed after this time. The delay duration and lithium concentration difference between the edge and center points are smaller for the cathode particles than for the anode particles. This discrepancy is caused by the difference between the diffusivity in the anode particles and cathode particles.

4.3. Electric potential distribution

Like the anisotropic distribution of the lithium concentration, the electric potential distribution has heterogeneous features. Fig. 7(a) and (b) show the electric potential distributions at

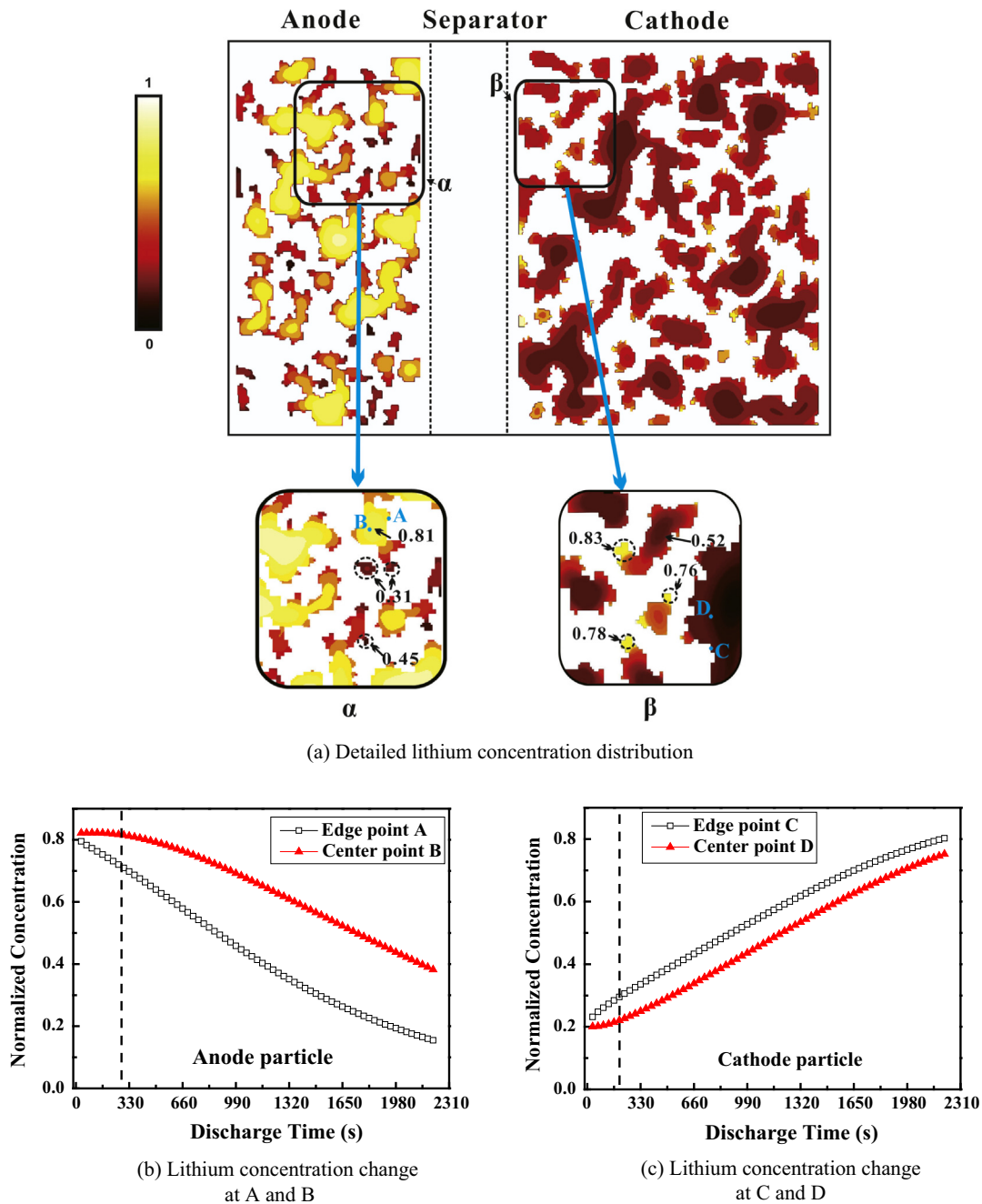


Fig. 6. Lithium concentration change in the particles.

different times and the corresponding lithium concentration distributions in the electrodes. As the discharge process continues, the electric potential in the anode area increases, whereas the electric potential in the cathode area decreases. The micro-scale electric potential changes in the anode and cathode particles can account for the macroscopic voltage drop during the discharge process. As shown in Fig. 7(a), two sections, i.e., α_1 in the cathode and α_2 in the anode, are selected to show the detailed lithium concentration distributions. Similarly, two sections, i.e., β_1 in the cathode and β_2 in the anode, are selected to show the corresponding electric potential distributions. In the electrode particles, the region with a lower lithium concentration produces a higher local electric potential, but the electric potential peaks occur at different locations for the cathode and anode particles. In the cathode par-

ticles, the area with a lower lithium concentration produces a higher local electric potential, and the electric potential peaks are located in the large particles, as shown in Fig. 7(c). As shown in Fig. 7(d), the area with a lower lithium concentration produces a higher electric potential in the anode particles. However, in the anode, the electric potential peaks occur in the small particles and at the corners and edges of the large particles. The local electric potential variation is induced by the lithium-dependent equilibrium potential. According to Eqs. (10) and (11), the change in the particle surface lithium concentration leads to a change in the equilibrium potential and finally induces the variation in the local electric potential. The remarkable microscopic local electric potential variation can compromise macroscopic performance. For instance, the local variations in charge-transfer kinetics and

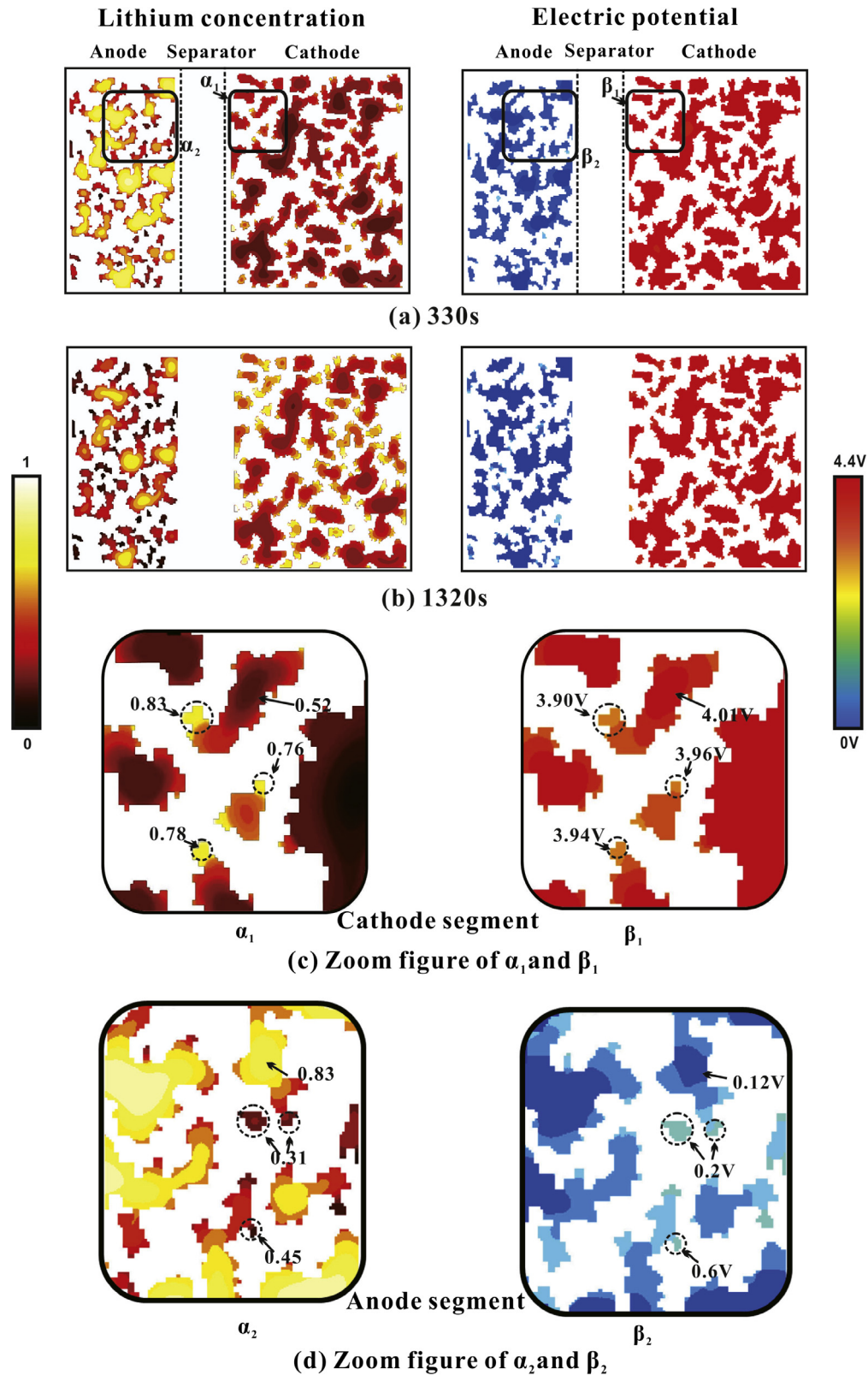


Fig. 7. Lithium concentrations and the corresponding electric potential distributions in the electrode particles at different time for the irregular electrode geometry.

possible solid electrolyte interface growth process may also occur depending on the local electrode voltages. Furthermore, excessive local voltage variations may also cause degradation, such as a local electrolyte breakdown.

4.4. Macroscopic discharge process for different particle sizes

The cell voltage response versus the depth of discharge (DOD) is obtained by using different electrode configurations to investigate

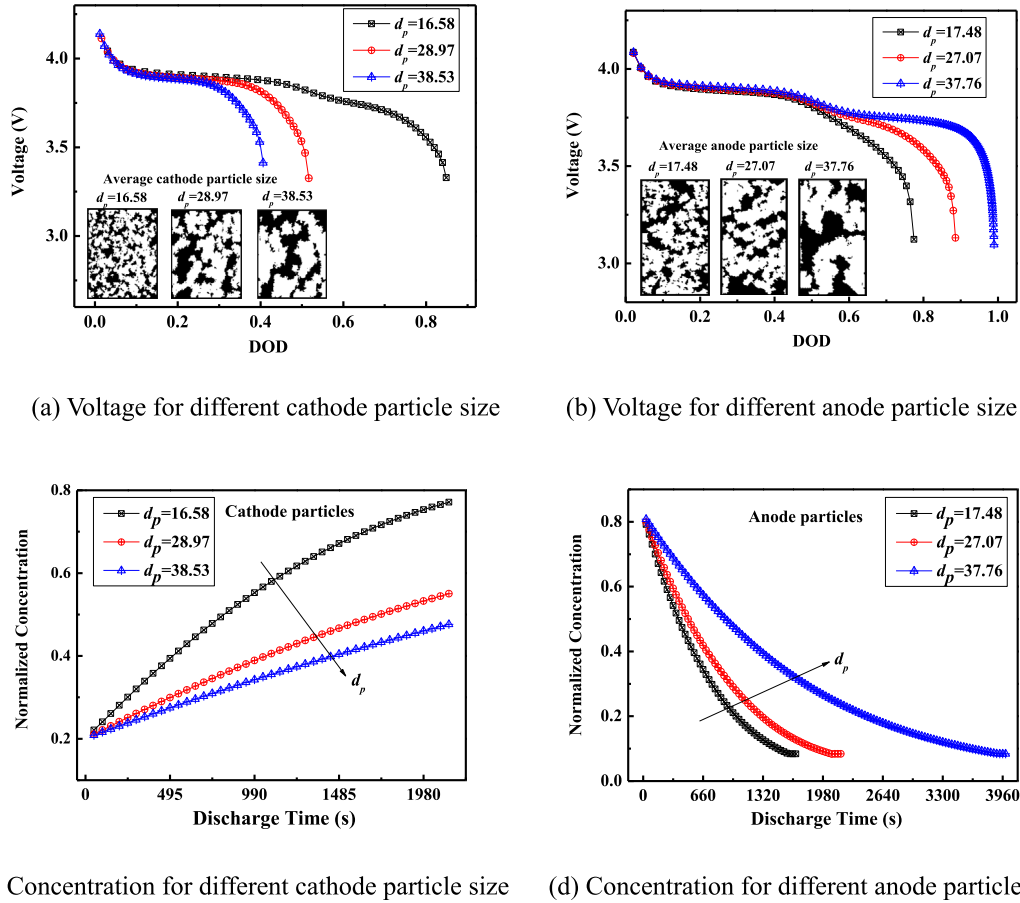


Fig. 8. Voltage responses of the electrodes with respect to the DOD and the corresponding average lithium concentrations in the particles with different particle sizes during the discharge process.

influence of the electrode microstructure on cell macroscopic performance. The state of charge (SOC) of the battery is assumed to be the SOC of the cathode. The SOC and the DOD are defined as follows:

$$SOC = \frac{(1/V_{e,s}) \int_{V_{e,s}} C_{Li} dV - C_{Li}^0}{C_{Li}^{\max} - C_{Li}^0} \quad (30)$$

$$DOD = 1 - SOC \quad (31)$$

where $V_{e,s}$ is the volume of the solid phase in the electrode, and $C_{Li}^{\max}$ and C_{Li}^0 are the average intercalated lithium concentrations at 100% and 0% SOC, respectively.

As reported in previous experiments [1,2], particle size is the parameter that influences macroscopic performance. Electrodes with different average particle sizes are generated to clarify the relationship between particle size and battery performance. Particle sizes, d_p , 16.58, 28.97, and 38.53 μm are selected for the cathode, and particle sizes, d_p , 17.48, 27.07, and 37.76 μm are selected for the anode. Fig. 8 shows the voltage response with respect to the DOD for the three different particle sizes in the anode and cathode. As shown in Fig. 8(a), the DOD of the cathode decreases significantly with increased particle diameter. In particular, the DOD decreases from 85% ($d_p = 16.58$ μm) to 40% ($d_p = 28.97$ μm). This result shows that better performance can be achieved by a smaller particle size in the cathode. In experiment studies [1,31] focus on lithium manganese oxide cathode and Ni-rich cathode, the similar conclusions have been reported that the discharge capacity is inversely proportional to the particle size.

However, Fig. 8(b) shows that superior performance is obtained with a larger particle size in the anode. The DOD reaches nearly 100% ($d_p = 37.36$ μm) but decreases to 77% ($d_p = 17.48$ μm).

The influence of the particle size on the discharge process is caused by the variation rate in the lithium concentration in the electrode particles. To quantify the lithium concentration change in the electrode particles, the corresponding normalized average lithium concentrations in the cathode and anode particles are depicted in Fig. 8(c) and (d), respectively. The lithium concentration in the cathode with a smaller particle size is higher than that in the cathode with a large particle size, as shown in Fig. 8(c). This result implies that more lithium ions diffuse into smaller cathode particles. For the anode, Fig. 8(d) shows that the lithium concentration in the anode particles decreases more slowly for a larger particle size, thereby providing more time for lithium to diffuse into the cathode particles. Consequently, a larger DOD can be achieved by a smaller cathode particle size and a larger anode size. The particles with different sizes are lithiated and delithiated at different rates, causing the change in the lithium-dependent equilibrium potential on the particle surface. The change in the lithium-dependent equilibrium potential finally affects the discharge performance.

4.5. Macroscopic discharge process for different electrode porosity values

The cell voltage responses of the electrode to different porosity values are shown in Fig. 9. The electrodes with three porosity values ($\varepsilon = 0.4, 0.5, 0.6$) are constructed for both the cathode and

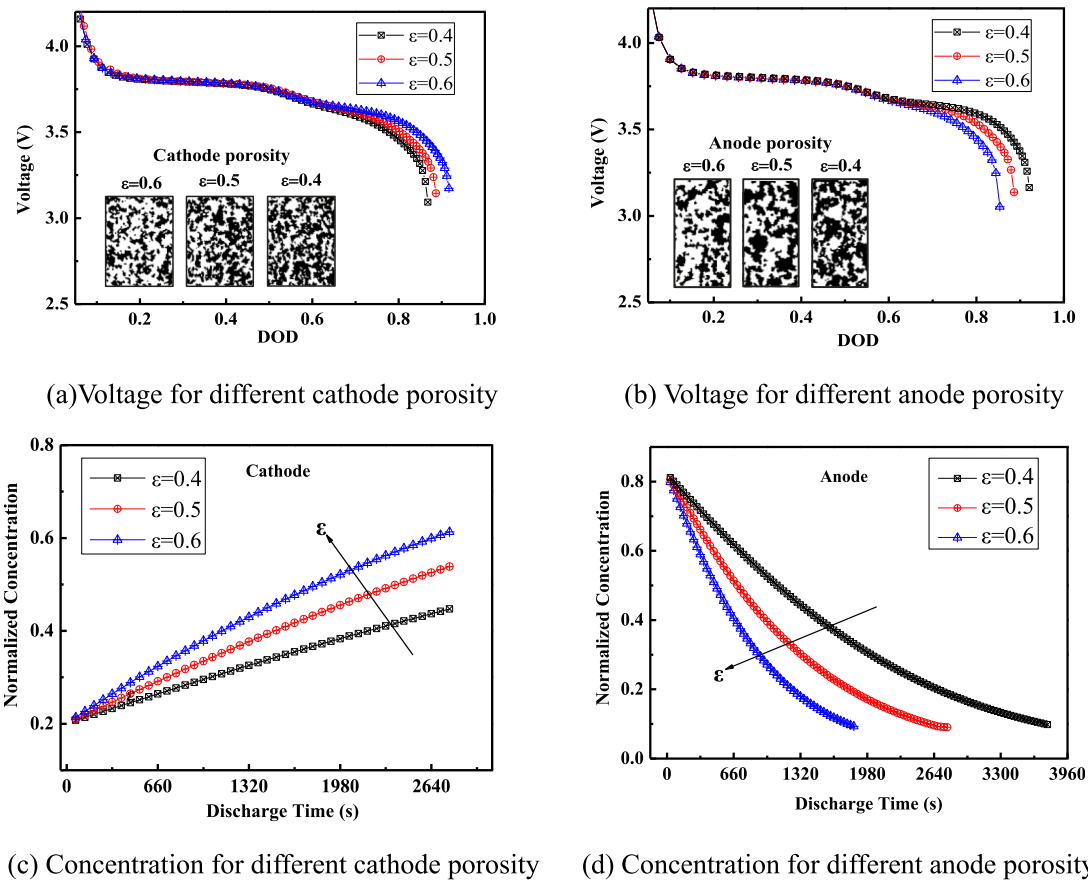


Fig. 9. Voltage responses of the electrodes with respect to the DOD and the corresponding average lithium concentrations in the particles with different electrode porosity values during the discharge process.

anode. As shown in Fig. 9(a) and (b), the predicted voltage responses in the cathode and anode are similar for different porosity values when the DOD is < 0.6 . By contrast, when the DOD is > 0.6 , the performance of the cathode increases as porosity increases, whereas an opposite tendency is observed for the anode. The electric potential is strongly influenced by the lithium concentration distribution in the cathode and anode. As shown in Fig. 9(c) and (d), the corresponding curves of the average lithium concentrations provide evidence that a higher porosity can achieve a higher lithium concentration for the cathode, whereas a lower porosity leads to a lower concentration drop rate for the anode.

5. Conclusion

A 2D lattice Boltzmann simulation scheme is proposed to investigate the influence of the morphology of electrode particles on the ion and electron transport in an LIB porous electrode during the discharge process. In the simulation, a $\text{Li}_y\text{C}_6|\text{Li}_x\text{Mn}_2\text{O}_4$ rocking-chair rechargeable battery structure is employed, and electrode geometries with different particle sizes and porosity values are generated. QSGS is utilized in the reconstruction of the electrode porous microstructure. The simulation results are qualitatively validated by comparing the voltage response of the LBM with that of the FEM-based model. The electrolyte lithium concentration in the cathode is low in the region located far from the separator and with a high particle density. High-lithium-concentration zones can be easily identified in smaller particles and at the edges and corners of large particles in the cathode. By contrast, low-lithium-concentration zones can be identified in small particles

and at the edges and corners of large particles in the anode. Moreover, a remarkable lag in the lithium concentration change occurs at the particle center point but not at the particle edge point. In the cathode and anode particles, the region with a lower lithium concentration produces a higher electric potential. The electric potential peaks occur in the large particles for the cathode but occur in the small particles and at the corners and edges of large particles for the anode. Larger anode and smaller cathode particles achieve superior performance. A decrease in the porosity of the anode and an increase in the porosity of the cathode induce a higher DOD. To overcome the performance decline of LIBs, the particles of the electrodes should be engineered such that they are monodispersed and uniformly distributed. In conclusion, the present study expects to promote the development of LIB investigations by presenting a new LBM modelling scheme and offering guidance to the porous electrode design.

Conflict of interest

We wish to confirm that there are no known conflicts of interest associated with this publication.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.ijheatmasstransfer.2018.03.004>.

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